

Grishma Dhakal

Computer Engineering Student | Pokhara, Nepal

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SUMMARY

Computer Engineering student dedicated to building robust software systems and exploring hardware-software integration. Driven by a passion for solving complex technical challenges and developing efficient, scalable applications.

EDUCATION

IOE Pashchimanchal Campus, Tribhuvan University Bachelor in Computer Engineering	Pokhara, Kaski, Nepal 2023 – Present
Capital Secondary School High School in Science; GPA: 3.68/4.0	Koteshwor, Kathmandu, Nepal 2021 – 2023
Kanti Secondary School School; GPA: 3.85/4.0	Butwal, Rupandehi, Nepal 2021

PROJECTS

voidend (Link)	Minor Project – Jun – Aug 2026
- Built a local-first, offline REST API simulation and mock data generation tool for frontend developers - Designed schema-driven endpoint configuration with realistic fake data generation using Faker.js and Zod - Implemented network simulation (latency, failure rate) and project-scoped JWT auth simulation	
shikshya (Link)	Hackathon – Feb 2026
- Built a full-stack educational platform using Next.js 15, TypeScript, tRPC, and PostgreSQL with Drizzle ORM - Integrated AI-powered learning assistant by extracting PDF content and injecting it as system context for Google Gemini via Vercel AI SDK - Engineered an interactive study workspace with resizable split-pane PDF viewer and real-time AI chat streaming	
rawhttp (Link)	Nov 2025
- Built a lightweight HTTP parser and server in Rust following RFC-9112 specifications for protocol parsing - Implemented thread-safe state management using Rust's ownership model and Arc primitives - Prevented request smuggling via duplicate Transfer-Encoding header detection and enforced header size limits	
notgit (Link)	Jul – Oct 2025
- Built a minimal reimplementation of Git in Go to understand version control systems under the hood - Learned the internal structure of Git objects (blobs, trees, commits) and storage mechanisms - Mastered CLI application architecture and command-line parsing logic	
g7-chat (Link)	Feb – May 2025
- Built a fast, minimalist AI chat interface with thread management and branching capabilities - Implemented AI integration allowing users to bring their own API keys with custom UI - Used tRPC to ensure end-to-end type safety across the full stack	
tube-and-fuss (Link)	Nov 2024
- Developed a backend system combining features of YouTube and Twitter using Express.js - Built REST API using JWT middleware for stateless authentication with modular architecture - Implemented MongoDB aggregation pipelines for efficient pagination and relational data fetching	

TECHNICAL SKILLS

Languages: TypeScript, JavaScript, Rust, Go, SQL, C/C++

Frameworks: Next.js, React, Svelte, Express, tRPC, Chi (Go), Tokio (Rust), Axum (Rust)

Databases: PostgreSQL, SQLite, MongoDB

Tools: Linux, Tmux, Vim/NeoVim, Git, Node.js, Docker, Vitest, pnpm

Libraries/ORM: Drizzle ORM, React Query, Tailwind CSS, Shadcn/UI, Zod, Better Auth, Vercel AI SDK